

# Cuaderno de Prácticas

MATEMÁTICAS DISCRETAS

Grado en Ingeniería Informática

**CURSO 2025/26**

---

## Datos personales

Nombre y apellidos : Francisco Javier Martín - Lunas Escobar

DNI : 26268082

Grupo de teoría : A

Grupo de prácticas : 4

## Códigos

```
In[1]:= (*
Capitulo 7 Producto cartesiano Ejemplo 7.8
*)
CARTESIANO[conjuntos_]:=Module[{i,j,k,cartesianotemp},cartesiano=Table[{conjuntos[[1,i]],{i,Length[conjuntos[[i,j]]]}],{i,Length[conjuntos]}];
Do[cartesianotemp={};
Do[AppendTo[cartesianotemp,Append[cartesiano[[k]],conjuntos[[i,j]]]];,{k,Length[cartesiano]}];
cartesiano=cartesianotemp;,{i,2,Length[conjuntos]}];
cartesiano];

(*
Capitulo 8 Relaciones binarias ejemplo 8.1
*)
AnalisisRB[A_,R_]:=Module[{},falloReflexiva={};
Do[If[MemberQ[R,{A[[n]],A[[n]]}],Null,AppendTo[falloReflexiva,A[[n]]]],{n,Length[A]}];
falloSimetrica={};
Do[If[MemberQ[R,{R[[m,2]],R[[m,1]]}],Null,AppendTo[falloSimetrica,{R[[m,2]],R[[m,1]]}]],{m,Length[R]}];
falloTransitiva={};
Do[Do[If[R[[p,1]]==R[[q,2]],If[MemberQ[R,{R[[q,1]],R[[p,2]]}],Null,AppendTo[falloTransitiva,{R[[q,1]],R[[p,2]]}]],{q,Length[R]}];
falloAntisimetrica={};
Do[If[MemberQ[R,{R[[r,2]],R[[r,1]]}]&&!(ToString[R[[r,1]]]==ToString[R[[r,2]]]),AppendTo[falloAntisimetrica,{R[[r,2]],R[[r,1]]}]],{r,Length[R]}];
If[falloReflexiva=={},Print["R es reflexiva"],Print["R no es reflexiva, falla en los elementos: "],Print[Union[falloReflexiva,A[[n]]]],{n,Length[A]}];
If[falloSimetrica=={},Print["R es simétrica"],Print["R no es simétrica, falla en los pares: "],Print[Union[falloSimetrica,{R[[m,2]],R[[m,1]]}]],{m,Length[R]}];
If[falloTransitiva=={},Print["R es transitiva"],Print["R no es transitiva, falla en los pares: "],Print[Union[falloTransitiva,{R[[p,1]],R[[p,2]]}]],{p,Length[R]}];
If[falloAntisimetrica=={},Print["R es antisimétrica"],Print["R no es antisimétrica, falla en los elementos: "],Print[Union[falloAntisimetrica,{R[[r,2]],R[[r,1]]}]],{r,Length[R]}];
If[Union[falloReflexiva,falloSimetrica,falloTransitiva]=={},Print["R es una relación de equivalencia"],Print["R no es una relación de equivalencia, falla en: "],Print[Union[falloReflexiva,falloSimetrica,falloTransitiva]]];
If[Union[falloReflexiva,falloAntisimetrica,falloTransitiva]=={},Print["R es una relación de orden"],Print["R no es una relación de orden, falla en: "],Print[Union[falloReflexiva,falloAntisimetrica,falloTransitiva]]];

(*
Capitulo 8 Relaciones de equivalencia Ejemplo 8.4
*)
COCIENTE[A_,R_]:=Module[{CONTADORi,CONTADORj,anadir},cociente={{A[[1]]}};
Do[anadir=True;
Do[If[Intersection[{A[[CONTADORi]],cociente[[CONTADORj]][[1]]}],R]!={},AppendTo[cociente[[CONTADORj]],A[[CONTADORi]]];anadir=False;
Break[;;]],{CONTADORj,1,Length[cociente]}];
If[anadir,AppendTo[cociente,{A[[CONTADORi]]}]],{CONTADORi,2,Length[A]}];
cociente];
```

```
In[4]:= (*
Capitulo 8 Relaciones de orden Ejemplo 8.9
*)
MAXIMALES[A_,R_]:=Module[{maximales,maximal,n,m},maximales={};
Do[maximal=True;
Do[If[MemberQ[R,{A[[n]],A[[m]]}]]&&n!=m,maximal=False;Break[;;]],{m,1,Length[A]}];
If[maximal,AppendTo[maximales,A[[n]]]],{n,1,Length[A]}];
maximales]
```

```
In[5]:= (*
Capitulo 8 Relaciones de orden Ejemplo 8.9
*)
MINIMALES[A_,R_] := Module[{minimales,minimal,n,m},minimales={};
Do[minimal=True;
Do[If[MemberQ[R,{A[[m]],A[[n]]}]&& n≠m,minimal=False;Break[;;]],{m,1,Length[A]}];
If[minimal,AppendTo[minimales,A[[n]]]],{n,1,Length[A]}];
minimales]
```

```
In[6]:= (*
Usar para calcular la imagen
Capitulo 7 Aplicaciones Ejemplo 7.16
*)
Imagen={};
Do[Imagen=Union[Imagen,Append[{} ,f[A[[i]]]],{i,1,Length[A]}];
Print["El conjunto imagen es: ",Imagen];
If[Length[Imagen]==Length[B],Print["Es sobreyectiva"],Print["No es sobreyectiva"]];
If[Length[A]==Length[Imagen],Print["Es inyectiva"],Print["No es inyectiva"]];
If[Length[A]==Length[B]&&Length[B]==Length[Imagen],Print["Es biyectiva"],Print["No es biyectiva"]];
]
```

## Práctica 6

### Ejercicio 7.9

Sea A el conjunto formado por todos los dígitos distintos de tu DNI. Calcular:

```
In[*]:= A79={2,6,8,0,"N"};
```

a)  $A \times A$

```
In[*]:= AxA=CARTESIANO[{A79,A79}]
```

```
Out[*]=
{{2, 2}, {6, 2}, {8, 2}, {0, 2}, {N, 2}, {2, 6}, {6, 6},
 {8, 6}, {0, 6}, {N, 6}, {2, 8}, {6, 8}, {8, 8}, {0, 8}, {N, 8}, {2, 0},
 {6, 0}, {8, 0}, {0, 0}, {N, 0}, {2, N}, {6, N}, {8, N}, {0, N}, {N, N}}
```

b)  $A \times A \times A$

```
In[*]:= AxAxA=CARTESIANO[{A79,A79,A79}]
```

```
Out[*]=
```

```
{ {2, 2, 2}, {6, 2, 2}, {8, 2, 2}, {0, 2, 2}, {N, 2, 2}, {2, 6, 2}, {6, 6, 2}, {8, 6, 2},
  {0, 6, 2}, {N, 6, 2}, {2, 8, 2}, {6, 8, 2}, {8, 8, 2}, {0, 8, 2}, {N, 8, 2}, {2, 0, 2},
  {6, 0, 2}, {8, 0, 2}, {0, 0, 2}, {N, 0, 2}, {2, N, 2}, {6, N, 2}, {8, N, 2}, {0, N, 2},
  {N, N, 2}, {2, 2, 6}, {6, 2, 6}, {8, 2, 6}, {0, 2, 6}, {N, 2, 6}, {2, 6, 6}, {6, 6, 6},
  {8, 6, 6}, {0, 6, 6}, {N, 6, 6}, {2, 8, 6}, {6, 8, 6}, {8, 8, 6}, {0, 8, 6}, {N, 8, 6},
  {2, 0, 6}, {6, 0, 6}, {8, 0, 6}, {0, 0, 6}, {N, 0, 6}, {2, N, 6}, {6, N, 6}, {8, N, 6},
  {0, N, 6}, {N, N, 6}, {2, 2, 8}, {6, 2, 8}, {8, 2, 8}, {0, 2, 8}, {N, 2, 8}, {2, 6, 8},
  {6, 6, 8}, {8, 6, 8}, {0, 6, 8}, {N, 6, 8}, {2, 8, 8}, {6, 8, 8}, {8, 8, 8}, {0, 8, 8},
  {N, 8, 8}, {2, 0, 8}, {6, 0, 8}, {8, 0, 8}, {0, 0, 8}, {N, 0, 8}, {2, N, 8}, {6, N, 8},
  {8, N, 8}, {0, N, 8}, {N, N, 8}, {2, 2, 0}, {6, 2, 0}, {8, 2, 0}, {0, 2, 0}, {N, 2, 0},
  {2, 6, 0}, {6, 6, 0}, {8, 6, 0}, {0, 6, 0}, {N, 6, 0}, {2, 8, 0}, {6, 8, 0}, {8, 8, 0},
  {0, 8, 0}, {N, 8, 0}, {2, 0, 0}, {6, 0, 0}, {8, 0, 0}, {0, 0, 0}, {N, 0, 0}, {2, N, 0},
  {6, N, 0}, {8, N, 0}, {0, N, 0}, {N, N, 0}, {2, 2, N}, {6, 2, N}, {8, 2, N}, {0, 2, N},
  {N, 2, N}, {2, 6, N}, {6, 6, N}, {8, 6, N}, {0, 6, N}, {N, 6, N}, {2, 8, N},
  {6, 8, N}, {8, 8, N}, {0, 8, N}, {N, 8, N}, {2, 0, N}, {6, 0, N}, {8, 0, N},
  {0, 0, N}, {N, 0, N}, {2, N, N}, {6, N, N}, {8, N, N}, {0, N, N}, {N, N, N}}
```

c) P(A)

```
In[*]:= PartesDeA=Subsets[A79]
```

```
Out[*]=
```

```
{ {}, {2}, {6}, {8}, {0}, {N}, {2, 6}, {2, 8}, {2, 0}, {2, N}, {6, 8}, {6, 0},
  {6, N}, {8, 0}, {8, N}, {0, N}, {2, 6, 8}, {2, 6, 0}, {2, 6, N}, {2, 8, 0},
  {2, 8, N}, {2, 0, N}, {6, 8, 0}, {6, 8, N}, {6, 0, N}, {8, 0, N}, {2, 6, 8, 0},
  {2, 6, 8, N}, {2, 6, 0, N}, {2, 8, 0, N}, {6, 8, 0, N}, {2, 6, 8, 0, N}}
```

d) Una partición de partes de AxA con tres elementos

```

In[*]:= A79d={6,2};
X=Subsets[CARTESIANO[{A79d,A79d}]]
A1={};
A2={};
A3={};
For[i=1, i<=Length[X],i++,
  If[Mod[i,3]==0,
    AppendTo[A1,X[[i]]]
  ];
  If[Mod[i,3]==1,
    AppendTo[A2,X[[i]]]
  ];
  If[Mod[i,3]==2,
    AppendTo[A3,X[[i]]]
  ];
]
W={A1,A2,A3}

```

```

Out[*]=
{{}, {{6, 6}}, {{2, 6}}, {{6, 2}}, {{2, 2}}, {{6, 6}, {2, 6}},
 {{6, 6}, {6, 2}}, {{6, 6}, {2, 2}}, {{2, 6}, {6, 2}}, {{2, 6}, {2, 2}},
 {{6, 2}, {2, 2}}, {{6, 6}, {2, 6}, {6, 2}}, {{6, 6}, {2, 6}, {2, 2}},
 {{6, 6}, {6, 2}, {2, 2}}, {{2, 6}, {6, 2}, {2, 2}}, {{6, 6}, {2, 6}, {6, 2}, {2, 2}}}

```

```

Out[*]=
{{{ {2, 6}}, {{6, 6}, {2, 6}}, {{2, 6}, {6, 2}}, {{6, 6}, {2, 6}, {6, 2}},
 {{2, 6}, {6, 2}, {2, 2}}}, {{}, {{6, 2}}, {{6, 6}, {6, 2}}, {{2, 6}, {2, 2}},
 {{6, 6}, {2, 6}, {2, 2}}, {{6, 6}, {2, 6}, {6, 2}, {2, 2}}},
 {{ {6, 6}}, {{2, 2}}, {{6, 6}, {2, 2}}, {{6, 2}, {2, 2}}, {{6, 6}, {6, 2}, {2, 2}}}}

```

Compruebo que ningún elemento de la partición es vacío

```

In[*]:= A1=={}
A2=={}
A3=={}

```

```

Out[*]=
False

```

```

Out[*]=
False

```

```

Out[*]=
False

```

Compruebo que son disjuntos 2 a 2

```
In[*]:= Intersection[A1,A2]=={}
Intersection[A1,A3]=={}
Intersection[A2,A3]=={}
```

```
Out[*]=
True
```

```
Out[*]=
True
```

```
Out[*]=
True
```

Compruebo que la unión da el total

```
In[*]:= SubsetQ[Union[A1,A2,A3],X]&&SubsetQ[X,Union[A1,A2,A3]]
```

```
Out[*]=
True
```

## Ejercicio 7.14

Sea A el conjunto formado por todas las letras distintas que aparecen en tu primer apellido y B el conjunto formado por todos los dígitos distintos de tu DNI, calcular si es posible:

```
In[12]:= A714={"M","A","R","T","I","N","L","U","S"};
B714={2,6,8,0,"N"};
```

a)  $A \times B$  y  $A \times A \times B$

```
In[*]:= CARTESIANO[{A714,B714}]
```

```
Out[*]=
{{M, 2}, {A, 2}, {R, 2}, {T, 2}, {I, 2}, {N, 2}, {L, 2}, {U, 2}, {S, 2},
 {M, 6}, {A, 6}, {R, 6}, {T, 6}, {I, 6}, {N, 6}, {L, 6}, {U, 6}, {S, 6},
 {M, 8}, {A, 8}, {R, 8}, {T, 8}, {I, 8}, {N, 8}, {L, 8}, {U, 8}, {S, 8},
 {M, 0}, {A, 0}, {R, 0}, {T, 0}, {I, 0}, {N, 0}, {L, 0}, {U, 0}, {S, 0},
 {M, N}, {A, N}, {R, N}, {T, N}, {I, N}, {N, N}, {L, N}, {U, N}, {S, N}}
```

```
In[*]:= CARTESIANO[{A714,A714,B714}]
```

```
Out[*]=
{{M, M, 2}, {A, M, 2}, {R, M, 2}, {T, M, 2}, {I, M, 2}, {N, M, 2}, {L, M, 2}, {U, M, 2},
 {S, M, 2}, {M, A, 2}, {A, A, 2}, {R, A, 2}, {T, A, 2}, {I, A, 2}, {N, A, 2}, {L, A, 2},
 {U, A, 2}, {S, A, 2}, {M, R, 2}, {A, R, 2}, {R, R, 2}, {T, R, 2}, {I, R, 2}, {N, R, 2},
 {L, R, 2}, {U, R, 2}, {S, R, 2}, {M, T, 2}, {A, T, 2}, {R, T, 2}, {T, T, 2}, {I, T, 2},
 {N, T, 2}, {L, T, 2}, {U, T, 2}, {S, T, 2}, {M, I, 2}, {A, I, 2}, {R, I, 2}, {T, I, 2},
 {I, I, 2}, {N, I, 2}, {L, I, 2}, {U, I, 2}, {S, I, 2}, {M, N, 2}, {A, N, 2}, {R, N, 2},
 {T, N, 2}, {I, N, 2}, {N, N, 2}, {L, N, 2}, {U, N, 2}, {S, N, 2}, {M, L, 2}, {A, L, 2},
 {R, L, 2}, {T, L, 2}, {I, L, 2}, {N, L, 2}, {L, L, 2}, {U, L, 2}, {S, L, 2}, {M, U, 2},
 {A, U, 2}, {R, U, 2}, {T, U, 2}, {I, U, 2}, {N, U, 2}, {L, U, 2}, {U, U, 2}, {S, U, 2},
 {M, S, 2}, {A, S, 2}, {R, S, 2}, {T, S, 2}, {I, S, 2}, {N, S, 2}, {L, S, 2}, {U, S, 2},
```

$\{S, S, 2\}, \{M, M, 6\}, \{A, M, 6\}, \{R, M, 6\}, \{T, M, 6\}, \{I, M, 6\}, \{N, M, 6\}, \{L, M, 6\},$   
 $\{U, M, 6\}, \{S, M, 6\}, \{M, A, 6\}, \{A, A, 6\}, \{R, A, 6\}, \{T, A, 6\}, \{I, A, 6\}, \{N, A, 6\},$   
 $\{L, A, 6\}, \{U, A, 6\}, \{S, A, 6\}, \{M, R, 6\}, \{A, R, 6\}, \{R, R, 6\}, \{T, R, 6\}, \{I, R, 6\},$   
 $\{N, R, 6\}, \{L, R, 6\}, \{U, R, 6\}, \{S, R, 6\}, \{M, T, 6\}, \{A, T, 6\}, \{R, T, 6\}, \{T, T, 6\},$   
 $\{I, T, 6\}, \{N, T, 6\}, \{L, T, 6\}, \{U, T, 6\}, \{S, T, 6\}, \{M, I, 6\}, \{A, I, 6\}, \{R, I, 6\},$   
 $\{T, I, 6\}, \{I, I, 6\}, \{N, I, 6\}, \{L, I, 6\}, \{U, I, 6\}, \{S, I, 6\}, \{M, N, 6\}, \{A, N, 6\},$   
 $\{R, N, 6\}, \{T, N, 6\}, \{I, N, 6\}, \{N, N, 6\}, \{L, N, 6\}, \{U, N, 6\}, \{S, N, 6\}, \{M, L, 6\},$   
 $\{A, L, 6\}, \{R, L, 6\}, \{T, L, 6\}, \{I, L, 6\}, \{N, L, 6\}, \{L, L, 6\}, \{U, L, 6\}, \{S, L, 6\},$   
 $\{M, U, 6\}, \{A, U, 6\}, \{R, U, 6\}, \{T, U, 6\}, \{I, U, 6\}, \{N, U, 6\}, \{L, U, 6\}, \{U, U, 6\},$   
 $\{S, U, 6\}, \{M, S, 6\}, \{A, S, 6\}, \{R, S, 6\}, \{T, S, 6\}, \{I, S, 6\}, \{N, S, 6\}, \{L, S, 6\},$   
 $\{U, S, 6\}, \{S, S, 6\}, \{M, M, 8\}, \{A, M, 8\}, \{R, M, 8\}, \{T, M, 8\}, \{I, M, 8\}, \{N, M, 8\},$   
 $\{L, M, 8\}, \{U, M, 8\}, \{S, M, 8\}, \{M, A, 8\}, \{A, A, 8\}, \{R, A, 8\}, \{T, A, 8\}, \{I, A, 8\},$   
 $\{N, A, 8\}, \{L, A, 8\}, \{U, A, 8\}, \{S, A, 8\}, \{M, R, 8\}, \{A, R, 8\}, \{R, R, 8\}, \{T, R, 8\},$   
 $\{I, R, 8\}, \{N, R, 8\}, \{L, R, 8\}, \{U, R, 8\}, \{S, R, 8\}, \{M, T, 8\}, \{A, T, 8\}, \{R, T, 8\},$   
 $\{T, T, 8\}, \{I, T, 8\}, \{N, T, 8\}, \{L, T, 8\}, \{U, T, 8\}, \{S, T, 8\}, \{M, I, 8\}, \{A, I, 8\},$   
 $\{R, I, 8\}, \{T, I, 8\}, \{I, I, 8\}, \{N, I, 8\}, \{L, I, 8\}, \{U, I, 8\}, \{S, I, 8\}, \{M, N, 8\},$   
 $\{A, N, 8\}, \{R, N, 8\}, \{T, N, 8\}, \{I, N, 8\}, \{N, N, 8\}, \{L, N, 8\}, \{U, N, 8\}, \{S, N, 8\},$   
 $\{M, L, 8\}, \{A, L, 8\}, \{R, L, 8\}, \{T, L, 8\}, \{I, L, 8\}, \{N, L, 8\}, \{L, L, 8\}, \{U, L, 8\},$   
 $\{S, L, 8\}, \{M, U, 8\}, \{A, U, 8\}, \{R, U, 8\}, \{T, U, 8\}, \{I, U, 8\}, \{N, U, 8\}, \{L, U, 8\},$   
 $\{U, U, 8\}, \{S, U, 8\}, \{M, S, 8\}, \{A, S, 8\}, \{R, S, 8\}, \{T, S, 8\}, \{I, S, 8\}, \{N, S, 8\},$   
 $\{L, S, 8\}, \{U, S, 8\}, \{S, S, 8\}, \{M, M, \emptyset\}, \{A, M, \emptyset\}, \{R, M, \emptyset\}, \{T, M, \emptyset\}, \{I, M, \emptyset\},$   
 $\{N, M, \emptyset\}, \{L, M, \emptyset\}, \{U, M, \emptyset\}, \{S, M, \emptyset\}, \{M, A, \emptyset\}, \{A, A, \emptyset\}, \{R, A, \emptyset\}, \{T, A, \emptyset\},$   
 $\{I, A, \emptyset\}, \{N, A, \emptyset\}, \{L, A, \emptyset\}, \{U, A, \emptyset\}, \{S, A, \emptyset\}, \{M, R, \emptyset\}, \{A, R, \emptyset\}, \{R, R, \emptyset\},$   
 $\{T, R, \emptyset\}, \{I, R, \emptyset\}, \{N, R, \emptyset\}, \{L, R, \emptyset\}, \{U, R, \emptyset\}, \{S, R, \emptyset\}, \{M, T, \emptyset\}, \{A, T, \emptyset\},$   
 $\{R, T, \emptyset\}, \{T, T, \emptyset\}, \{I, T, \emptyset\}, \{N, T, \emptyset\}, \{L, T, \emptyset\}, \{U, T, \emptyset\}, \{S, T, \emptyset\}, \{M, I, \emptyset\},$   
 $\{A, I, \emptyset\}, \{R, I, \emptyset\}, \{T, I, \emptyset\}, \{I, I, \emptyset\}, \{N, I, \emptyset\}, \{L, I, \emptyset\}, \{U, I, \emptyset\}, \{S, I, \emptyset\},$   
 $\{M, N, \emptyset\}, \{A, N, \emptyset\}, \{R, N, \emptyset\}, \{T, N, \emptyset\}, \{I, N, \emptyset\}, \{N, N, \emptyset\}, \{L, N, \emptyset\}, \{U, N, \emptyset\},$   
 $\{S, N, \emptyset\}, \{M, L, \emptyset\}, \{A, L, \emptyset\}, \{R, L, \emptyset\}, \{T, L, \emptyset\}, \{I, L, \emptyset\}, \{N, L, \emptyset\}, \{L, L, \emptyset\},$   
 $\{U, L, \emptyset\}, \{S, L, \emptyset\}, \{M, U, \emptyset\}, \{A, U, \emptyset\}, \{R, U, \emptyset\}, \{T, U, \emptyset\}, \{I, U, \emptyset\}, \{N, U, \emptyset\},$   
 $\{L, U, \emptyset\}, \{U, U, \emptyset\}, \{S, U, \emptyset\}, \{M, S, \emptyset\}, \{A, S, \emptyset\}, \{R, S, \emptyset\}, \{T, S, \emptyset\}, \{I, S, \emptyset\},$   
 $\{N, S, \emptyset\}, \{L, S, \emptyset\}, \{U, S, \emptyset\}, \{S, S, \emptyset\}, \{M, M, N\}, \{A, M, N\}, \{R, M, N\}, \{T, M, N\},$   
 $\{I, M, N\}, \{N, M, N\}, \{L, M, N\}, \{U, M, N\}, \{S, M, N\}, \{M, A, N\}, \{A, A, N\}, \{R, A, N\},$   
 $\{T, A, N\}, \{I, A, N\}, \{N, A, N\}, \{L, A, N\}, \{U, A, N\}, \{S, A, N\}, \{M, R, N\}, \{A, R, N\},$   
 $\{R, R, N\}, \{T, R, N\}, \{I, R, N\}, \{N, R, N\}, \{L, R, N\}, \{U, R, N\}, \{S, R, N\}, \{M, T, N\},$   
 $\{A, T, N\}, \{R, T, N\}, \{T, T, N\}, \{I, T, N\}, \{N, T, N\}, \{L, T, N\}, \{U, T, N\}, \{S, T, N\},$   
 $\{M, I, N\}, \{A, I, N\}, \{R, I, N\}, \{T, I, N\}, \{I, I, N\}, \{N, I, N\}, \{L, I, N\}, \{U, I, N\},$   
 $\{S, I, N\}, \{M, N, N\}, \{A, N, N\}, \{R, N, N\}, \{T, N, N\}, \{I, N, N\}, \{N, N, N\}, \{L, N, N\},$   
 $\{U, N, N\}, \{S, N, N\}, \{M, L, N\}, \{A, L, N\}, \{R, L, N\}, \{T, L, N\}, \{I, L, N\}, \{N, L, N\},$   
 $\{L, L, N\}, \{U, L, N\}, \{S, L, N\}, \{M, U, N\}, \{A, U, N\}, \{R, U, N\}, \{T, U, N\},$   
 $\{I, U, N\}, \{N, U, N\}, \{L, U, N\}, \{U, U, N\}, \{S, U, N\}, \{M, S, N\}, \{A, S, N\},$   
 $\{R, S, N\}, \{T, S, N\}, \{I, S, N\}, \{N, S, N\}, \{L, S, N\}, \{U, S, N\}, \{S, S, N\}$

b) Una aplicación entre A y P(B) que sea inyectiva y otra entre P(B) y A que sea sobreyectiva.

```

In[*]:= Dominio=A714;
Codominio=Subsets[B714];
If[Length[Dominio]≤Length[Codominio], (*Si el dominio tiene menos elementos, puedo definir un
    For[i=1, i≤Length[Dominio], i++,
        f[Dominio[[i]]=Codominio[[i]];
    ]
    ,
    Print["No se puede definir una aplicación inyectiva"];
]
A=Dominio;
B=Codominio;
Imagen={};
Do[Imagen=Union[Imagen,Append[{},f[A[[i]]]],{i,1,Length[A]}];
Print["El conjunto imagen es: ",Imagen];
If[Length[Imagen]==Length[B],Print["Es sobreyectiva"],Print["No es sobreyectiva"]];
If[Length[A]==Length[Imagen],Print["Es inyectiva"],Print["No es inyectiva"]];
If[Length[A]==Length[B]&&Length[B]==Length[Imagen],Print["Es biyectiva"],Print["No es biyectiva"]];

```

El conjunto imagen es: {{}, {0}, {2}, {6}, {8}, {N}, {2, 0}, {2, 6}, {2, 8}}

No es sobreyectiva

Es inyectiva

No es biyectiva

```

In[*]:= Clear;

```

```

In[14]:= Dominio=Subsets[B714];
Codominio=A714;
If[Length[Dominio]≥Length[Codominio], (*Si el dominio tiene más elementos, puedo definir una
    For[i=1, i≤Length[Dominio], i++,
        f[Dominio[[i]]=Codominio[[Mod[i,Length[Codominio]]+1]]; (*ojito índice del codominio cuando
    ]
    ,
    Print["No se puede definir una aplicación Sobreyectiva"];
]
A=Dominio;
B=Codominio;
Imagen={};
Do[Imagen=Union[Imagen,Append[{},f[A[[i]]]],{i,1,Length[A]}];
Print["El conjunto imagen es: ",Imagen];
If[Length[Imagen]==Length[B],Print["Es sobreyectiva"],Print["No es sobreyectiva"]];
If[Length[A]==Length[Imagen],Print["Es inyectiva"],Print["No es inyectiva"]];
If[Length[A]==Length[B]&&Length[B]==Length[Imagen],Print["Es biyectiva"],Print["No es biyectiva"]];

```

El conjunto imagen es: {A, I, L, M, N, R, S, T, U}

Es sobreyectiva

No es inyectiva

No es biyectiva

c) Una aplicación no inyectiva de  $A \times B$  en  $P(A \times B)$



```

In[ ]:= Dominio=CARTESIANO[{A714,B714}];
c=CARTESIANO[{A714,B714}];
Codominio=Subsets[c];
For[i=1, i≤Length[Dominio], i++,
    f[Dominio[[i]]=Codominio[[1]];
]
A=Dominio;
B=Codominio;
Imagen={};
Do[Imagen=Union[Imagen,Append[{},f[A[[i]]]],{i,1,Length[A]}];
Print["El conjunto imagen es: ",Imagen];
If[Length[Imagen]==Length[B],Print["Es sobreyectiva"],Print["No es sobreyectiva"]];
If[Length[A]==Length[Imagen],Print["Es inyectiva"],Print["No es inyectiva"]];
If[Length[A]==Length[B]&&Length[B]==Length[Imagen],Print["Es biyectiva"],Print["No es biyectiva"]]

```

General: The current computation was aborted because there was insufficient memory available to complete the computation.

Throw: Uncaught SystemException returned to top level. Can be caught with Catch[...,\_SystemException].

Out[ ]:=

SystemException[MemoryAllocationFailure]

Part: Part specification Codominio[[1]] is longer than depth of object. [i](#)

El conjunto imagen es: {Codominio[[1]]}

No es sobreyectiva

No es inyectiva

No es biyectiva

d) Una aplicación no sobreyectiva de  $P(A \times B)$  en  $A \times B$

```

In[ ]:= Dominio=Subsets[CARTESIANO[{X,Y}]];
Codominio=CARTESIANO[{X,Y}];
For[i=1, i≤Length[Dominio], i++,
    f[Dominio[[i]]=Codominio[[1]];
]
A=Dominio;
B=Codominio;
Imagen={};
Do[Imagen=Union[Imagen,Append[{},f[A[[i]]]],{i,1,Length[A]}];
Print["El conjunto imagen es: ",Imagen];
If[Length[Imagen]==Length[B],Print["Es sobreyectiva"],Print["No es sobreyectiva"]];
If[Length[A]==Length[Imagen],Print["Es inyectiva"],Print["No es inyectiva"]];
If[Length[A]==Length[B]&&Length[B]==Length[Imagen],Print["Es biyectiva"],Print["No es biyectiva"]]

```

Part: Part 1 of {} does not exist. [i](#)

El conjunto imagen es: {{}}[[1]]

No es sobreyectiva

Es inyectiva

No es biyectiva

## Ejercicio 18 Rel 2

Sean  $A=\{a,b,c\}$ ,  $B=\{1,2,3\}$ ,  $A_1=\{a,b\}$  y  $B_1=\{2,3\}$ . Definir, si existe, una aplicación biyectiva  $f:A \times P(B_1) \rightarrow P(A_1) \times B$  y hacer todas las comprobaciones explícitamente

```
In[*]:= A18={"a","b","c"};
B18={1,2,3};
A181={"a","b"};
B181={2,3};
Dominio=CARTESIANO[{A18,Subsets[B181]}];
Codominio=CARTESIANO[{Subsets[A181],B18}];
Dominio
Codominio
A=Dominio;
B=Codominio;
Imagen={};
Do[Imagen=Union[Imagen,Append[{},f[A[[i]]]],{i,1,Length[A]}];
Print["El conjunto imagen es: ",Imagen];
If[Length[Imagen]==Length[B],Print["Es sobreyectiva"],Print["No es sobreyectiva"]];
If[Length[A]==Length[Imagen],Print["Es inyectiva"],Print["No es inyectiva"]];
If[Length[A]==Length[B]&&Length[B]==Length[Imagen],Print["Es biyectiva"],Print["No es biyectiva"]];
```

```
Out[*]=
{{a, {}}, {b, {}}, {c, {}}, {a, {2}}, {b, {2}}, {c, {2}},
 {a, {3}}, {b, {3}}, {c, {3}}, {a, {2, 3}}, {b, {2, 3}}, {c, {2, 3}}}
```

```
Out[*]=
{{{ }, 1}, {{a}, 1}, {{b}, 1}, {{a, b}, 1}, {{ }, 2}, {{a}, 2},
 {{b}, 2}, {{a, b}, 2}, {{ }, 3}, {{a}, 3}, {{b}, 3}, {{a, b}, 3}}

El conjunto imagen es: {{{ }, 1}, {{ }, 2}, {{ }, 3}, {{a}, 1}, {{a}, 2},
 {{a}, 3}, {{b}, 1}, {{b}, 2}, {{b}, 3}, {{a, b}, 1}, {{a, b}, 2}, {{a, b}, 3}}

Es sobreyectiva
Es inyectiva
Es biyectiva
```

```
In[*]:= If[Length[Dominio]==Length[Codominio], (*Compruebo que dominio y codominio tienen los mismos e
For[i=1, i<=Length[Dominio], i++,
f[Dominio[[i]]]=Codominio[[i]];
]
,
Print["No se puede definir una aplicación biyectiva"];
]
```

## Práctica 7

### Ejercicio 8.1

Sea  $A=\{a,b,c,d,e\}$ , consideramos la relación binaria  $R=\{(a,b),(b,c),(d,e)\}$

In[175]:=

```
A81={"a","b","c","d","e"};
```

a) Completar la relación binaria R para que sea una relación de orden total

In[176]:=

```
R81a={
{"a","a"}, {"a","b"}, {"a","c"}, {"a","d"}, {"a","e"},
{"b","b"}, {"b","c"}, {"b","d"}, {"b","e"},
{"c","c"}, {"c","e"}, {"c","d"},
{"d","d"}, {"d","e"},
{"e","e"}
};
AnalisisRB[A81,R81a]
```

R es reflexiva

R no es simétrica, falla en los pares:

```
{{b, a}, {c, a}, {d, a}, {e, a}, {c, b}, {d, b}, {e, b}, {e, c}, {d, c}, {e, d}}
```

R es transitiva

R es antisimétrica

R no es relación de equivalencia

R es una relación de orden

In[178]:=

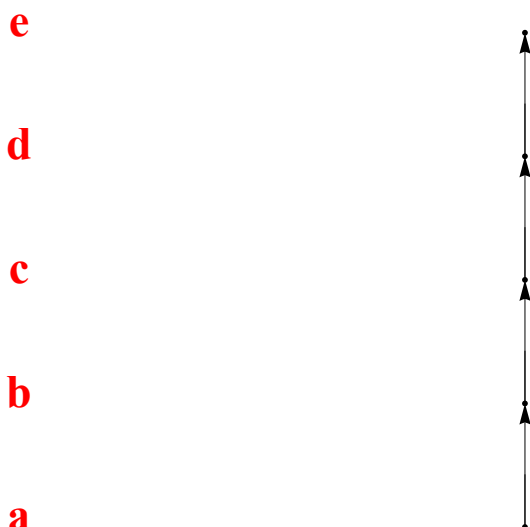
```

A=A81;
R=R81a;
Clear[Coord];
tabla=Table[0,{i1,Length[A]},{j1,3}];
B=A;
t1=1;
nivel=0;
While[B!={},minimales={};nivel++;
Do[minimal=True;
Do[If[Intersection[{B[[m1]],B[[n1]]},R]!={},{m1,n1},minimal=False];,{m1,1,Length[B]}];
If[minimal,AppendTo[minimales,B[[n1]]];
tabla[[t1]]={nivel,B[[n1]],0};
t1++;];,{n1,1,Length[B]}];
B=Complement[B,minimales];];
R1={};
Do[AppendTo[R1,{A[[i1]],A[[i1]]},{i1,1,Length[A]}];
R=Complement[R,R1];
R1={};
Do[Do[If[Length[Intersection[R,{R[[k1,1]],A[[j1]]},{A[[j1]],R[[k1,2]]}]]==2,R1=Union[R1,{R[[k1]]}];];
R=Complement[R,R1];
puntos={};t1=0;
Do[cont=0;
Do[If[tabla[[i1,1]]==j1,cont=cont+1},{i1,1,Length[A]}];
Do[t1++;
puntos=Union[puntos,{Text[Style[tabla[[t1,2]],Large,Bold,Red,Background->None,FontFamily->Times],
tabla[[t1,3]]-k1-(cont/2);,{k1,1,cont}],{j1,1,tabla[[Length[A],1]]}];
Coord[elem_]:=Do[If[elem==tabla[[h1,2]],Coord[elem]={tabla[[h1,3]],tabla[[h1,1]]},{h1,1,Length[A]}];
Do[Coord[A[[i1]]},{i1,1,Length[A]}];
Do[AppendTo[puntos,Arrow[{Coord[R[[t1,1]]],Coord[R[[t1,2]]]}];,{t1,1,Length[R]}];
Print["Diagrama de orden:"]
Show[Graphics[puntos],AspectRatio->1,Background->None]

```

Diagrama de orden:

Out[198]=



Como se observa en el diagrama de Hasse el diagrama de orden es de una relación de orden total

b) Completar la relación binaria R para que sea reflexiva y transitiva, pero no sea simétrica y antisimétrica

```
In[*]:= R81b={
{"a","a"}, {"a","b"}, {"a","c"}, {"a","d"}, {"a","e"},
{"b","b"}, {"b","c"}, {"b","d"}, {"b","e"}, {"b","a"},
{"c","c"}, {"c","e"}, {"c","d"},
{"d","d"}, {"d","e"},
{"e","e"}
};
AnalisisRB[A81,R81b]
```

R es reflexiva

R no es simétrica, falla en los pares:

$\{\{c, a\}, \{d, a\}, \{e, a\}, \{c, b\}, \{d, b\}, \{e, b\}, \{e, c\}, \{d, c\}, \{e, d\}\}$

R es transitiva

R no es antisimétrica, falla en los pares:  $\{\{a, b\}, \{b, a\}\}$

R no es relación de equivalencia

R no es relación de orden

## Ejercicio 8.8

En el conjunto  $A=\{1,2,3,4\}$  se establece la relación binaria

$R=\{\{1,1\},\{2,2\},\{2,3\},\{2,4\},\{3,2\},\{3,3\},\{3,4\},\{4,2\},\{4,3\},\{4,4\}\};$

Justificar que es una relación de equivalencia y calcular el conjunto cociente. Comprobar que dicho conjunto de una partición de A.

```
In[*]:= A88={1,2,3,4};
R88={ {1,1}, {2,2}, {2,3}, {2,4}, {3,2}, {3,3}, {3,4}, {4,2}, {4,3}, {4,4} };
AnalisisRB[A88,R88]
```

R es reflexiva

R es simétrica

R es transitiva

R no es antisimétrica, falla en los pares:  $\{\{2, 3\}, \{2, 4\}, \{3, 2\}, \{3, 4\}, \{4, 2\}, \{4, 3\}\}$

R es una relación de equivalencia

R no es relación de orden

```
In[*]:= COCIENTE[A88,R88]
```

```
Out[*]=
{{1}, {2, 3, 4}}
```

Demostración de que A es una partición

```
In[*]:= A1={1};
A2={2,3,4};
A1≠{} && A2≠{} (*ningun elemento de la particion es vacio*)
Intersection[A1,A2]=={} (*son disjuntos dos a dos, si hay mas elementos en la particion hay qu
Union[A1,A2]==A88 (*La unión de todos da el total*)
```

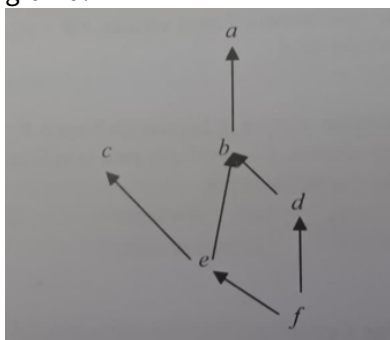
```
Out[*]=
True
```

```
Out[*]=
True
```

```
Out[*]=
True
```

## Ejercicio 8.14

Sea  $X=\{a,b,c,d,e,f\}$  junto con la ordenación dada por el siguiente diagrama:



Sea  $Y=\{b,f,e\}$  y  $Z=\{e,b,c,a\}$  subconjunto de  $X$ . Se pide:

```
In[46]:= X814={"a","b","c","d","e","f"};
R814={{ "f","f"}, {"f","e"}, {"f","d"}, {"f","c"}, {"f","b"}, {"f","a"}, {"e","e"}, {"e","c"}, {"e","b"}, {"e","a"}, {"d","d"}, {"d","c"}, {"d","b"}, {"d","a"}, {"c","c"}, {"c","b"}, {"c","a"}, {"b","b"}, {"b","a"}, {"a","a"};
Y814={"b","f","e"};
Z814={"e","b","c","a"};
```

a) Elementos máximos y minimales de  $X, Y, Z$ .

```
In[*]:= MAXIMALES [X814,R814]
MINIMALES [X814,R814]
```

```
Out[*]=
{a, c}
```

```
Out[*]=
{f}
```

```
In[*]:= MAXIMALES [Y814,R814]
MINIMALES [Y814,R814]
```

```
Out[*]=
{b}
```

```
Out[*]=
{f}
```

```
In[*]:= MAXIMALES [Z814,R814]
MINIMALES [Z814,R814]
```

```
Out[*]=
{c, a}
```

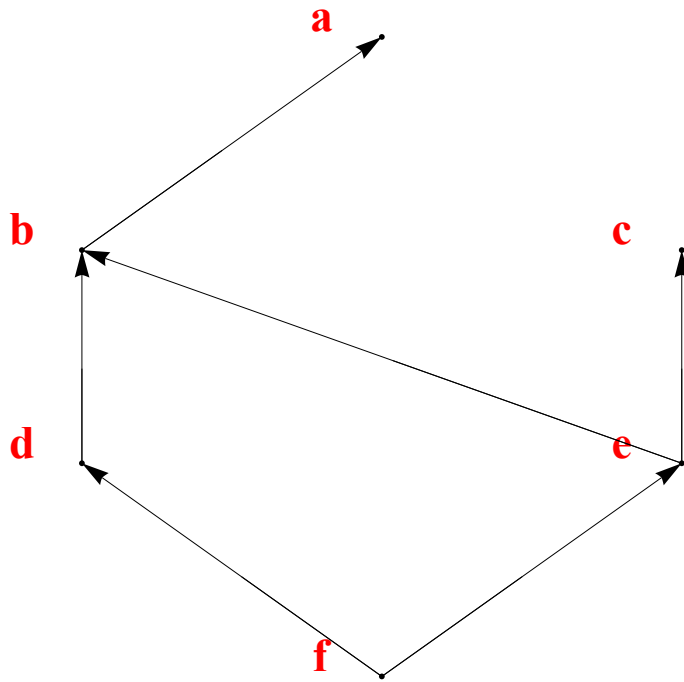
```
Out[*]=
{e}
```

b) Representa con Mathematica el diagrama de orden de X, Y y Z

```
In[50]:= A=X814;
R=R814;
Clear[Coord];
tabla=Table[0,{i1,Length[A]},{j1,3}];
B=A;
t1=1;
nivel=0;
While[B!= {},minimales={};nivel++;
Do[minimal=True;
Do[If[Intersection[{B[[m1]],B[[n1]]},{R}]=={}&& n1!=m1,minimal=False];,{m1,1,Length[B]}];
If[minimal,AppendTo[minimales,B[[n1]]];
tabla[[t1]]={nivel,B[[n1]],0};
t1++;];,{n1,1,Length[B]}];
B=Complement[B,minimales];];
R1={};
Do[AppendTo[R1,{A[[i1]],A[[i1]]},{i1,1,Length[A]}];
R=Complement[R,R1];
R1={};
Do[Do[If[Length[Intersection[R,{R[[k1,1]],A[[j1]]},{A[[j1]],R[[k1,2]]}]]==2,R1=Union[R1,{R[[k1]]}];];
R=Complement[R,R1];
puntos={};t1=0;
Do[cont=0;
Do[If[tabla[[i1,1]]==j1,cont=cont+1},{i1,1,Length[A]}];
Do[t1++;
puntos=Union[puntos,{Text[Style[tabla[[t1,2]],Large,Bold,Red,Background->None,FontFamily->Times],
tabla[[t1,3]]-k1-(cont/2);,{k1,1,cont}],{j1,1,tabla[[Length[A],1]]}];
Coord[elem_]:=Do[If[elem==tabla[[h1,2]],Coord[elem]={tabla[[h1,3]],tabla[[h1,1]]},{h1,1,Length[A]}];
Do[Coord[A[[i1]]},{i1,1,Length[A]}];
Do[AppendTo[puntos,Arrow[{Coord[R[[t1,1]]],Coord[R[[t1,2]]]}];,{t1,1,Length[R]}];
Print["Diagrama de orden:"]
Show[Graphics[puntos],AspectRatio->1,Background->None]
```

Diagrama de orden:

Out[70]=





In[133]:=

```

A=Y814;
R={{ "f", "f"}, {"f", "e"}, {"f", "b"}, {"e", "e"}, {"e", "b"}, {"b", "b"} };
Clear[Coord];
tabla=Table[0, {i1, Length[A]}, {j1, 3}];
B=A;
t1=1;
nivel=0;
While[B!= {}, minimales={}; nivel++;
Do[minimal=True;
Do[If[Intersection[{B[[m1]], B[[n1]]}, R] != {} && n1 != m1, minimal=False];, {m1, 1, Length[B]}];
If[minimal, AppendTo[minimales, B[[n1]]];
tabla[[t1]]={nivel, B[[n1]], 0};
t1++;];, {n1, 1, Length[B]}];
B=Complement[B, minimales];];
R1={};
Do[AppendTo[R1, {A[[i1]], A[[i1]]}], {i1, 1, Length[A]}];
R=Complement[R, R1];
R1={};
Do[Do[If[Length[Intersection[R, {{R[[k1, 1]], A[[j1]]}, {A[[j1]], R[[k1, 2]]}}]] == 2, R1=Union[R1, {R[[k1]]}];];
R=Complement[R, R1];
puntos={}; t1=0;
Do[cont=0;
Do[If[tabla[[i1, 1]] == j1, cont=cont+1], {i1, 1, Length[A]}];
Do[t1++;
puntos=Union[puntos, {Text[Style[tabla[[t1, 2]], Large, Bold, Red, Background->None, FontFamily->Times],
tabla[[t1, 3]] = k1 - (cont/2);, {k1, 1, cont}], {j1, 1, tabla[[Length[A], 1]]}];
Coord[elem_] := Do[If[elem == tabla[[h1, 2]], Coord[elem] = {tabla[[h1, 3]], tabla[[h1, 1]]}, {h1, 1, Length[A]}];
Do[Coord[A[[i1]]], {i1, 1, Length[A]}];
Do[AppendTo[puntos, Arrow[{Coord[R[[t1, 1]]], Coord[R[[t1, 2]]]}];, {t1, 1, Length[R]}];
Print["Diagrama de orden:"]
Show[Graphics[puntos], AspectRatio->1, Background->None]

```

Diagrama de orden:

Out[153]=

**b****e****f**

In[154]:=

```

A=Z814;
R={{{"e","e"}, {"e","c"}, {"e","b"}, {"e","a"}, {"b","b"}, {"b","a"}, {"a","a"}, {"c","c"}}};
Clear[Coord];
tabla=Table[0,{i1,Length[A]},{j1,3}];
B=A;
t1=1;
nivel=0;
While[B!= {},minimales={};nivel++;
Do[minimal=True;
Do[If[Intersection[{B[[m1]],B[[n1]]},R]!={}&& n1!=m1,minimal=False];,{m1,1,Length[B]}];
If[minimal,AppendTo[minimales,B[[n1]]];
tabla[[t1]]={nivel,B[[n1]],0};
t1++;];,{n1,1,Length[B]}];
B=Complement[B,minimales];];
R1={};
Do[AppendTo[R1,{A[[i1]],A[[i1]]},{i1,1,Length[A]}];
R=Complement[R,R1];
R1={};
Do[Do[If[Length[Intersection[R,{R[[k1,1]],A[[j1]]},{A[[j1]],R[[k1,2]]}]]==2,R1=Union[R1,{R[[k1]]}];];
R=Complement[R,R1];
puntos={};t1=0;
Do[cont=0;
Do[If[tabla[[i1,1]]==j1,cont=cont+1},{i1,1,Length[A]}];
Do[t1++;
puntos=Union[puntos,{Text[Style[tabla[[t1,2]],Large,Bold,Red,Background->None,FontFamily->Times],
tabla[[t1,3]]-k1-(cont/2);,{k1,1,cont}],{j1,1,tabla[[Length[A],1]]}];
Coord[elem_]:=Do[If[elem==tabla[[h1,2]],Coord[elem]={tabla[[h1,3]],tabla[[h1,1]]},{h1,1,Length[A]}];
Do[Coord[A[[i1]]},{i1,1,Length[A]}];
Do[AppendTo[puntos,Arrow[{Coord[R[[t1,1]],Coord[R[[t1,2]]}]]];,{t1,1,Length[R]}];
Print["Diagrama de orden:"]
Show[Graphics[puntos],AspectRatio->1,Background->None]

```

Diagrama de orden:

Out[174]=

